

Syllabus prescribed by
PONDICHERRY UNIVERSITY
(w.e.f. 2014 – 2015)

MA T41 MATHEMATICS – IV

UNIT – I

PARTIAL DIFFERENTIAL EQUATIONS : Formation by elimination of arbitrary constants and arbitrary functions – General, singular, particular and integrals – Lagrange's linear first order equation – Higher order differential equations with constant coefficients

UNIT – II

PARTIAL DIFFERENTIAL EQUATIONS : Solution of partial differential equation by the method of separation of variables – Boundary value problems – Fourier series solution – Transverse vibration of an elastic string.

UNIT – III

PARTIAL DIFFERENTIAL EQUATIONS : Fourier series solution for one dimensional heat flow equation – Fourier series solutions for two dimensional heat flow equations under steady state condition – (Cartesian and Polar forms).

UNIT – IV

APPLIED STATISTICS : Curve fitting by the method of least squares – fitting of straight lines, second degree parabolas and more general curves. Test of significance: Large samples test for single proportions, differences of proportions, single mean, difference of means and standard deviations.

UNIT – V

APPLIED STATISTICS : Small samples – Test for single mean, difference of means and correlations of coefficients, test for ratio of variances – Chi-square test for goodness of fit and independence of attributes.

Unit - 5

Applied Statistics

Type - I Chi-Square Test of Goodness of Fit

$$\chi^2 = \sum \left[\frac{(O - E)^2}{E} \right]$$

where $O \rightarrow$ observed frequency

$E \rightarrow$ Expected frequency

degrees of freedom $\gamma = n - 1$

Note: The degrees of freedom for

fitting a Binomial distribution is $n - 1$

fitting a Poisson distribution is $n - 2$

fitting a normal distribution is $n - 3$

- ①. The table below gives the no. of aircraft accidents that occurred during the various days of the week. Test whether the accidents are uniformly distributed over the week.

Day of the week :	Mon	Tue	Wed	Thur	Fri	Sat
No. of accidents :	14	18	12	11	15	14

Soln:

Null hypothesis (H_0): Accidents are uniformly distributed over the week.

The expected frequencies of the accidents on each of these days are

$$\frac{14 + 18 + 12 + 11 + 15 + 14}{6} = \frac{84}{6} = 14$$

O $\rightarrow$ observed frequency

E $\rightarrow$ Expected frequency

O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
14	14	0	0
18	14	16	1.14286
12	14	4	0.28571
11	14	9	0.64286
15	14	1	0.07143
14	14	0	0
			2.14286

calculated value of $\chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$

$$= 2.14286$$

Table value of χ^2 for $\gamma = n-1$ $(\because n=6)$

$$= 6 - 1 = 5 \text{ degrees}$$

of freedom at 5% level of Significance = 11.07.

$$\because \alpha = 5\%$$

$$= \frac{5}{100} = 0.05$$

$$d.f = n-1$$

$$= 6-1=5$$

Since, calculated value of

$$\chi^2 < \text{table value of } \chi^2$$

$\therefore$ Accept H_0 .

Hence, the accidents are uniformly distributed over the week.

2. In 120 throws of a single die, the following distribution of faces were obtained.

Faces :	1	2	3	4	5	6	Total
Frequency :	30	25	18	10	22	15	120

compute the statistic you would use to test whether the results constituted replications of equal probability, we use chi-square test.

Soln:

Null hypothesis (H_0): Die is unbiased.

The expected frequency for each face

1 to 6 are

$$\frac{30 + 25 + 18 + 10 + 22 + 15}{6} = \frac{120}{6} = 20$$

O $\rightarrow$ observed frequency

E $\rightarrow$ Expected frequency

O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
30	20	100	5.00
25	20	25	1.25
18	20	4	0.20
10	20	100	5.00
22	20	4	0.20
15	20	25	1.25
			12.90

$\therefore$ calculated value of $\chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$

$$= 12.90$$

Table value of χ^2

$$\text{degrees of freedom } \gamma = n - 1 \\ = 6 - 1 = 5$$

$$\alpha = 5\% = \frac{5}{100} = 0.05$$

$$\text{Table value} = 11.07$$

Since,

calculated value of $\chi^2 > \text{table value of } \chi^2$

$\therefore$ Reject H_0 .

Hence the die is biased.

3. A die is thrown 276 times and the results of these throws are given below,

No. application the die :	1	2	3	4	5	6
Frequency :	40	32	29	59	57	59
Test whether the die is biased or not.						

Soln: $[H_0: \text{Die is unbiased}]$

4. A sample analysis of examination results of 500 students was made. It was found that 220 students have failed, 170 have secured a third class, 90 have secured a second class and the rest of the students are first class. Do these figures support the general belief that the above categories are in the ratio $4:3:2:1$ respectively?

Soln:

Null hypothesis (H_0): The result categories are in the ratio $4:3:2:1$

The expected frequencies are given in the ratio $4:3:2:1$. Since the total number of students are 500. Now we divide the total students 500 corresponding to the

$$4:3:2:1 \text{ are } 500 \times \frac{4}{10},$$

$$500 \times \frac{3}{10}, 500 \times \frac{2}{10}, 500 \times \frac{1}{10}.$$

ie. 200, 150, 100, 50.

Let $O \rightarrow$ observed frequency

$E \rightarrow$ Expected frequency

Result status	O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
Failures	220	200	400	2.0
Third class	170	150	400	2.6667
Second class	90	100	100	1.000
First class	20	50	900	18.000
	<u>500</u>			22.6667

$\therefore$ calculated value of $\chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$

$= 22.6667$

degrees of freedom $\gamma = n - 1$ [$\because n = 4$]
 $= 4 - 1 = 3$

$\alpha = 5\% = \frac{5}{100} = 0.05$

$\Rightarrow 7.82$

Since, calculated value of $\chi^2 >$ table value of χ^2 .

$\therefore$ Reject H_0 .

Hence the four categories are not in the ratio $4:3:2:1$.

5. A set of 5 identical coins is tossed 320 times and the number of heads appearing each time is recorded:

No. of heads :	0	1	2	3	4	5
Frequency :	14	45	80	112	61	8

Test whether the coins are unbiased at 5% level.

Soln:

Null hypothesis (H_0): Coin is unbiased.

That is $p = P(\text{get head}) = \frac{1}{2}$

$$\therefore q = \frac{1}{2}$$

$$\begin{aligned} [q &= 1 - p] \\ &= 1 - \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

We use the binomial distribution to calculate the theoretical frequency. Using the binomial distribution we have,

$$P(r) = {}^n C_r p^r q^{n-r}$$

$$\text{Sub } n=5, p=\frac{1}{2}, q=\frac{1}{2}$$

$$P(r) = {}^5 C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{5-r}$$

$$= {}^5 C_r \left(\frac{1}{2}\right)^{5-r+r} = {}^5 C_r \left(\frac{1}{2}\right)^5$$

$$r = 0, 1, 2, 3, 4, 5$$

Let $N(r)$ = No. of times getting r heads
 $= N \times P(X=r)$.

where N is the total no. of tosses = 320

$$N(0) = N \times P(0) = 320 \times {}^5C_0 \left(\frac{1}{2}\right)^5 = 10$$

$[\because {}^5C_0 = 1]$

$$N(1) = N \times P(1) = 320 \times {}^5C_1 \left(\frac{1}{2}\right)^5 = 50$$

$$N(2) = N \times P(2) = 320 \times {}^5C_2 \left(\frac{1}{2}\right)^5 = 100$$

$$N(3) = N \times P(3) = 320 \times {}^5C_3 \left(\frac{1}{2}\right)^5 = 100$$

$$N(4) = N \times P(4) = 320 \times {}^5C_4 \left(\frac{1}{2}\right)^5 = 50$$

$$N(5) = N \times P(5) = 320 \times {}^5C_5 \left(\frac{1}{2}\right)^5 = 10$$

$$[\because {}^5C_1 = \frac{5}{1}$$

$${}^5C_2 = \frac{5 \times 4}{1 \times 2} = 10$$

$${}^5C_3 = \frac{5 \times 4 \times 3}{1 \times 2 \times 3} = 10$$

$${}^5C_4 = \frac{5 \times 4 \times 3 \times 2}{1 \times 2 \times 3 \times 4} = 5$$

$${}^5C_5 = \frac{5 \times 4 \times 3 \times 2 \times 1}{1 \times 2 \times 3 \times 4 \times 5} = 1]$$

Let $O \rightarrow$ Observed frequency

$E \rightarrow$ Expected frequency

No. of heads	O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
0	14	10	16	1.60
1	45	50	25	0.50
2	80	100	400	4.00
3	112	100	144	1.44
4	61	50	121	2.42
5	8	10	4	0.40
	<u>320</u>			<u>10.36</u>

$$\therefore \text{calculated value of } \chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$$

$$= 10.36$$

$$\text{degree of freedom } \} = n-1$$

$$= 6-1 = 5$$

$$\alpha = 5\% = \frac{5}{100} = 0.05$$

table value $\Rightarrow 11.07$

Since,

$$\text{calculated value of } \chi^2 < \text{table value of } \chi^2$$

$\therefore$ Accept H_0 .

Thus coin is unbiased.

6. Five coins are tossed 256 times. The number of heads observed is given below.

No. of heads (x): 0 1 2 3 4 5

Frequency (f): 5 35 75 84 45 12

Examine if the coin is true, by applying chi-square test of goodness of fit.

Soln:

Null hypothesis (H_0): Coin is unbiased.

That is $p = P(\text{get head}) = \frac{1}{2}$.

$$\therefore q = \frac{1}{2} \quad [q = 1 - p]$$

We use the binomial distribution to calculate the theoretical frequency. Using the binomial distribution we have,

$$P(r) = {}^n C_r p^r q^{n-r} \quad \text{sub } n=5, p=\frac{1}{2}, q=\frac{1}{2}.$$

$$= {}^5 C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{5-r}$$

$$= {}^5 C_r \left(\frac{1}{2}\right)^5, \quad r = 0, 1, 2, 3, 4, 5.$$

$$\begin{aligned} \text{Let } N(r) &= \text{No. of times getting } r \text{ heads} \\ &= N \times P(x=r). \end{aligned}$$

where N is the total no. of tosses
 $= 256$

$$N(0) = N \times P(0) = 256 \times 5C_0 \left(\frac{1}{2}\right)^5 = 8$$

$$N(1) = N \times P(1) = 256 \times 5C_1 \left(\frac{1}{2}\right)^5 = 40$$

$$N(2) = N \times P(2) = 256 \times 5C_2 \left(\frac{1}{2}\right)^5 = 80$$

$$N(3) = N \times P(3) = 256 \times 5C_3 \left(\frac{1}{2}\right)^5 = 80$$

$$N(4) = N \times P(4) = 256 \times 5C_4 \left(\frac{1}{2}\right)^5 = 40$$

$$N(5) = N \times P(5) = 256 \times 5C_5 \left(\frac{1}{2}\right)^5 = 8$$

Let $O \rightarrow$ Observed frequency

$E \rightarrow$ Expected frequency

No. of Heads	O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
0	5	8	9	1.125
1	35	40	25	0.625
2	75	80	25	0.313
3	84	80	16	0.200
4	45	40	25	0.625
5	12	8	16	2.000
	256			4.888

$$\therefore \text{calculated value of } \chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$$

$$= 4.888$$

Find Table value degrees of freedom $\gamma = n-1 = 5$

$$\alpha = 5\% = 0.05$$

$$\text{Table value} = 11.07$$

Since, calculated value of $\chi^2 < \text{table value of } \chi^2$.

$\therefore$ Accept H_0 .

Thus the coin is unbiased.

7. Fit a poisson distribution to the following data and also test the good ness of fit:

x	0	1	2	3	4	5	Total
y	142	156	69	27	5	1	400

Soln: Null hypothesis (H_0): Fit a poisson

distribution is a good ness of fit to the given data.

w.k.t the Poisson distribution with parameter λ is given by

$$P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}, \quad r=0,1,2,3,4,5$$

To find λ :

x	0	1	2	3	4	5	Total
f	142	156	69	27	5	1	400
$f \cdot x$	0	156	138	81	20	5	400

$$\lambda = \bar{x} = \frac{\sum fx}{\sum f} = \frac{400}{400} = 1$$

The expected frequencies are given by

$$N(r) = N \times P(r) = N \times \frac{e^{-\lambda} \lambda^r}{r!}$$

$$= 400 \times \frac{e^{-1} (1)^r}{r!} = 400 \times \frac{e^{-1}}{r!}$$

$r=0,1,2,3,4,5$

$$r=0,1,2,3,4,5$$

$N(r)$ is the number of pages containing

r errors, and $N = \sum f = 400$.

$$N(0) = 400 \times \frac{e^{-1}}{0!} = 147.152$$

$$N(1) = 400 \times \frac{e^{-1}}{1!} = 147.152$$

$$N(2) = 400 \times \frac{e^{-1}}{2!} = 73.576$$

$$N(3) = 400 \times \frac{e^{-1}}{3!} = 24.525$$

$$N(4) = 400 \times \frac{e^{-1}}{4!} = 6.131$$

$$N(5) = 400 \times \frac{e^{-1}}{5!} = 1.226$$

Let $O \rightarrow$ observed frequency

$E \rightarrow$ Expected frequency

x	O	E	$O-E$	$(O-E)^2$	$\frac{(O-E)^2}{E}$
✓ 0	142	147	-5	25	0.1701
✓ 1	156	147	9	81	0.5510
✓ 2	69	74	-5	25	0.3378
✓ 3	27	25	2	4	0.16
✓ 4	5	6	-1	1	0.1429
✓ 5	1	1	0	0	0
					<u>1.3619</u>

Since the above last two frequencies are small - so we pool them together.

$$\chi^2 = \sum \left[\frac{(O-E)^2}{E} \right] = 1.3619$$

$$\text{degrees of freedom} = n - 2 = 5 - 2 = 3$$

$$\text{table value} = 7.815$$

Since, calculated value $<$ table value

$\therefore$ Accept H_0 .

Thus, a fitting of a poisson distribution to the good news of fit of the given data is satisfactory.

$\frac{f_i}{\sum f_i}$	$\frac{f_i}{\sum f_i}$	$\frac{f_i}{\sum f_i}$	$\frac{f_i}{\sum f_i}$	$\frac{f_i}{\sum f_i}$	$\frac{f_i}{\sum f_i}$
0.0500	0.0500	0.0500	0.0500	0.0500	0.0500
0.1000	0.1000	0.1000	0.1000	0.1000	0.1000
0.2000	0.2000	0.2000	0.2000	0.2000	0.2000
0.3000	0.3000	0.3000	0.3000	0.3000	0.3000
0.4000	0.4000	0.4000	0.4000	0.4000	0.4000
0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
0.6000	0.6000	0.6000	0.6000	0.6000	0.6000
0.7000	0.7000	0.7000	0.7000	0.7000	0.7000
0.8000	0.8000	0.8000	0.8000	0.8000	0.8000
0.9000	0.9000	0.9000	0.9000	0.9000	0.9000
1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

20. The number of accidents occurring in a factory during a year is given below. Test the hypothesis that the number of accidents follows a poisson distribution.

$$E(x) = \frac{\sum f_i x_i}{\sum f_i} = \frac{100}{20} = 5$$

$\therefore$ All the k values of f_i are

$$f_i = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$f_i = \frac{e^{-5} 5^k}{k!}$$

Example 5.5.16. When the first proof of 392 pages of a book of 1200 pages were read, the distribution of printing mistakes were found to be as follows:

Number of mistakes in a page (x) :	0	1	2	3	4	5	6	
Number of pages (f) :	275	72	30	7	5	2	1	Nov.'09

Fit a Poisson distribution to the above data and test goodness of fit.

Solution . Null hypothesis (H_0) : Fit a Poisson distribution is a good ness of fit to the given data.

W.k.t the Poisson distribution with parameter λ is given by

$$P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}, \quad r = 0, 1, 2, \dots$$

To find λ :

$x :$	0	1	2	3	4	5	6	Total
$f :$	275	72	30	7	5	2	1	392
$fx :$	0	72	60	21	20	10	6	189

$$\lambda = \bar{x} = \frac{\sum fx}{\sum f} = \frac{189}{392} = 0.4821$$

The expected frequencies are given by

$$\begin{aligned} N(r) &= N \times P(r) = N \times \frac{e^{-\lambda} \lambda^r}{r!} \\ &= 392 \times \frac{e^{-0.4821} (0.4821)^r}{r!}, \quad r = 0, 1, 2, 3, 4, 5, 6 \end{aligned}$$

$N(r)$ is the number of pages containing r errors and $N = \sum f = 392$.

$$N(0) = 392 \times \frac{e^{-0.4821} (0.4821)^0}{0!} = 242.054$$

$$N(1) = 392 \times \frac{e^{-0.4821} (0.4821)^1}{1!} = 116.694$$

$$N(2) = 392 \times \frac{e^{-0.4821} (0.4821)^2}{2!} = 28.129$$

$$N(3) = 392 \times \frac{e^{-0.4821} (0.4821)^3}{3!} = 4.520$$

$$N(4) = 392 \times \frac{e^{-0.4821} (0.4821)^4}{4!} = 0.545$$

$$N(5) = 392 \times \frac{e^{-0.4821} (0.4821)^5}{5!} = 0.0525$$

$$N(6) = 392 \times \frac{e^{-0.4821} (0.4821)^6}{6!} = 0.00422$$

Let O be the observed frequency and E be the expected frequency.

x	0	1	2	3	4	5	6
O	275	72	30	7	5	2	1
				15			
E	242.1	116.7	28.1	4.5	0.5	0.1	0
				5.1			
$O - E$	32.9	-44.7	1.9	15 - 5.1 = 9.9			
$(O - E)^2$	1082.41	1998.09	3.61	98.01			
$\frac{(O-E)^2}{E}$	4.4709	17.1216	0.1285	19.217			

[Adjust $\Sigma E = \Sigma O$.] Since the above last four frequencies are small. So we pool them together.

$$\chi^2 = \sum \left[\frac{(O - E)^2}{E} \right] = 4.4709 + 17.121 + 0.1285 + 19.217$$

$$= 40.938$$

$$\therefore \text{Calculated value of } \chi^2 = \sum \left[\frac{(O - E)^2}{E} \right] = 40.938$$

Table value of χ^2 for $\nu = n - 2 = 4 - 2 = 2$ degrees of freedom at 5% level of significance = 5.99

Since, calculated value of $\chi^2 >$ table value of χ^2 . $\therefore$ Reject H_0 . Thus, fitting Poisson for the given distribution is not satisfactory.

Example 5.5.17. The number of defects per unit in a sample of 330 units of a manufactured products was found as follows:

Number of defects : 0 1 2 3 4

Number of pages (f) : 214 92 20 3 1

Fit a Poisson distribution and test the goodness of fit.

Solution . Null hypothesis (H_0) : Fit a Poisson distribution is a good ness of fit to the given data.

W.k.t the probability law of Poisson distribution is

$$P(X = r) = \frac{e^{-\lambda} \lambda^r}{r!}, \quad r = 0, 1, 2, \dots$$

Where λ is a parameter (or) mean of the Poisson distribution.

To find λ :

$x :$	0	1	2	3	4	Total
$f :$	214	92	20	3	1	330
$fx :$	0	92	40	9	4	145

$$\lambda = \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{145}{330} = 0.44$$

The expected frequencies are given by

$$\begin{aligned} N(r) &= N \times P(r) \\ &= N \times \frac{e^{-\lambda} \lambda^r}{r!} \\ &= 330 \times \frac{e^{-0.44} (0.44)^r}{r!}, \quad r = 0, 1, 2, 3, 4 \end{aligned}$$

$N(r)$ is the number of pages containing r defects and $N = \Sigma f = 330$.

$$N(0) = 330 \times \frac{e^{-0.44} (0.44)^0}{0!} = 212.65$$

$$N(1) = 330 \times \frac{e^{-0.44} (0.44)^1}{1!} = 93.57$$

$$N(2) = 330 \times \frac{e^{-0.44} (0.44)^2}{2!} = 20.58$$

$$N(3) = 330 \times \frac{e^{-0.44} (0.44)^3}{3!} = 3.02$$

$$N(4) = 330 \times \frac{e^{-0.44} (0.44)^4}{4!} = 0.33$$

Let O be the observed frequency and E be the expected frequency.

x	0	1	2	3	4
O	214	92	20	3	1
			24		
E	1.35	-1.57	20.58	3.02	0.33
			23.93		
$O - E$	1.8225	2.4649	$24 - 23.93 = 0.07$		
$(O - E)^2$	1.8225	2.4649	0.049		
$\frac{(O - E)^2}{E}$	0.009	0.026	0.0002		

Since the above last three frequencies are small. So we pool them together.

$$\begin{aligned}
 \chi^2 &= \sum \left[\frac{(O - E)^2}{E} \right] \\
 &= 0.009 + 0.026 + 0.0002 \\
 &= 0.035
 \end{aligned}$$

$$\therefore \text{Calculated value of } \chi^2 = \sum \left[\frac{(O - E)^2}{E} \right] = 0.035$$

Table value of χ^2 for $\nu = n - 2 = 3 - 2 = 1$ degrees of freedom at 5% level of significance = 3.84

Since, calculated value of $\chi^2 >$ table value of χ^2 . $\therefore$ Reject H_0 . Thus, fitting Poisson for the given distribution is not satisfactory.

Example 5.5.18. For the following distribution a Poisson fit was made and the expected frequencies are given with the observed frequencies:

Value x	0	1	2	4	5	6
Frequencies f	142	156	69	27	5	1
Expected frequencies f_e	147	147	74	25	6	1

χ^2 test to test Independence of Attributes

$$\chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$$

$p \rightarrow$ row
 $q \rightarrow$ column

degrees of freedom $\gamma = (p-1)(q-1)$

Note:

The value of χ^2 test for 2×2 contingency table can be calculated as follows.

Table of Observed Frequencies:

	A_1	A_2	total
B_1	a	b	$a+b$
B_2	c	d	$c+d$
	$a+c$	$b+d$	N

where $N = a+b+c+d$.

Table of Expected Frequencies:

	A_1	A_2
B_1	$\frac{(a+b)(a+c)}{N}$	$\frac{(a+b)(b+d)}{N}$
B_2	$\frac{(a+c)(c+d)}{N}$	$\frac{(b+d)(c+d)}{N}$

1. The following data is collected on two characters based on this, can you say that there is no relation between smoking and literacy?

	Smokers	Non-smokers
Literates	83	57
Illiterates	45	68

Soln:

Null hypothesis (H_0): Literacy and smoking habit are independent.

Given observed frequencies are

	Smokers	Non-smokers	Total
Literates	83	57	140
Illiterates	45	68	113
Total	128	125	253

$$O_1 = 83, O_2 = 57,$$

$$O_3 = 45, O_4 = 68$$

Then expected frequencies are

	Smokers	Non-Smokers	total
Literates	$E_1 = \frac{128 \times 140}{253} = 70.83$	$E_2 = \frac{125 \times 140}{253} = 69.17$	140
Illiterates	$E_3 = \frac{128 \times 113}{253} = 57.17$	$E_4 = \frac{125 \times 113}{253} = 55.83$	113
total	128	125	253

O	E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
83	70.83	148.11	2.09
57	69.17	148.11	2.14
45	57.17	148.11	2.59
68	55.83	148.11	2.65
			9.47

$\therefore$ calculated value $\chi^2 = \sum \left(\frac{(O-E)^2}{E} \right)$

$= 9.47$

degrees of freedom $y = (p-1)(q-1)$

$= (2-1)(2-1) = 1$

$\alpha = 5\% = 0.05$

table value = 3.84

p → row
q → column

Since, calculated value $>$ table value

$\therefore$ Reject H_0 .

Thus, the literacy and smoking habits are dependent.

2. Two researchers A and B adopted different techniques while rating the students level. Can you say that the techniques adopted by them are significant?

Researchers	Below Ave.	Average	Above Ave.	Genius	Total
A	40	33	25	2	100
B	86	60	44	10	200
Total	126	93	69	12	300

Soln: Null hypothesis (H_0):

Two attributes are independent.

Let the observed frequencies are

$$O_1 = 40, O_2 = 33, O_3 = 25, O_4 = 2, O_5 = 86$$

$$O_7 = 44 \text{ and } O_8 = 60.$$

Then the expected frequencies are

$E_1, E_2, E_3, E_4, E_5, E_6, E_7, E_8$.

Researchers	Below Average	Average	Above Average	Genius	Total
A	$\frac{100 \times 126}{300} = 42$	$\frac{100 \times 93}{300} = 31$	$\frac{100 \times 69}{300} = 23$	$\frac{100 \times 12}{300} = 4$	100
B	$\frac{200 \times 126}{300} = 84$	$\frac{200 \times 93}{300} = 62$	$\frac{200 \times 69}{300} = 46$	$\frac{200 \times 12}{300} = 8$	200
Total	126	93	69	12	300

O	E	(O-E)	(O-E) ²	$\frac{(O-E)^2}{E}$
40	42	-2	4	0.0952
33	31	2	4	0.129
25	23	2	4	0.1739
2	4	-2	4	1
86	84	2	4	0.0476
60	62	-2	4	0.0645
44	46	-2	4	0.087
10	8	2	4	0.5
				2.0972

$\therefore$ calculated value of $\chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$

$$= 2.0972 //$$

$$\text{degrees of freedom } \chi^2 = (p-1)(q-1)$$

$$= (4-1)(2-1) = 3(1) = 3$$

$$\alpha = 5\% \quad \alpha = 0.05$$

$$\text{Table value} = 7.815$$

Since, calculated value of $\chi^2 < \text{table value of } \chi^2$

$\therefore$ Accept H_0 .

Thus no significant difference between the technique of the researchers.

3. A total number of 3759 individuals were interviewed in a public opinion survey on a political proposal. Of them 1872 were men and rest women. A total of 2257 individuals were in favour of the proposal and 917 were opposed to it. A total of 243 men were undecided and 442 women were opposed to the proposal. Do you justify or contradict the hypothesis that there is no association between sex and attitude?

Soln : H_0 : There is no association between sex and attitude.

The given observed frequencies are
 [where the underlined values are determined by
 ourself from the given values].

Persons	Favoured	Opposed	Undecided	total
Men	<u>1154</u> ^②	<u>475</u> ^①	243	1872
women	<u>1103</u> ^③	442	<u>342</u> ^⑤	<u>1887</u> ^④
Total	2257	917	<u>585</u> ^⑥	3759

The table for expected frequencies are given by

Persons	Favoured	Opposed	Undecided	total
Men	$\frac{1872 \times 2257}{3759} = 1124$	$\frac{1872 \times 917}{3759} = 457$	$\frac{1872 \times 585}{3759} = 291$	1872
Women	$\frac{1887 \times 2257}{3759} = 1133$	$\frac{1887 \times 917}{3759} = 460$	$\frac{1887 \times 585}{3759} = 294$	1887
total	2257	917	585	3759

Let $O \rightarrow$ observed frequency

$E \rightarrow$ Expected frequency

O	E	(O-E)	(O-E) ²	$\frac{(O-E)^2}{E}$
1154	1124	30	900	0.801
475	457	18	324	0.710
243	291	-48	2304	7.916
1103	1133	-30	900	0.794
442	460	-18	324	0.704
342	294	48	2304	7.837
				18.762

$\therefore$ calculated value of $\chi^2 = \sum \left[\frac{(O-E)^2}{E} \right]$
 $= 18.762$

degrees of freedom = $(p-1)(q-1)$
 $= (2-1)(3-1) = 1 \cdot 2 = 2$

$\alpha = 5\% = 0.05$

table value = 5.99

Since, calculated value of $\chi^2 >$ table value of χ^2

$\therefore$ Reject H_0 .

Thus, there is some association between
 Sex and attitude.

Type-I 't' test for single Mean

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

Working rule:

$[\because n(<30)]$

1. Write the sample size (n), sample mean ($\bar{x}$), Population mean (μ), sample S.D (s).

2. State Null hypothesis H_0 and alternative hypothesis H_1 .

3. Find the calculated value of z by using test statistic $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$.

and consider $|t|$ if t becomes negative.

4. Write tabulated value of t for $r = n-1$ degrees of freedom at 5% level of significance.

5. conclusion.

(i). If calculated value of $t <$ table value of t , then accept H_0 .

(or)
(ii). If calculated value of $t >$ table value of t , then reject H_0 .

1. Ten oil tins are taken from an automatic filling machine. The mean weight of the tins is 15.8 kg and standard deviation 0.5 kg.

Does the sample mean differ significantly

from the intended weight 16 kg.

Soln:

Given $n = 10 (< 30)$, $\bar{x} = 15.8$

$\mu = 16$ Sample S.D.S = 0.5

Null hypothesis (H_0) : $\mu = 16$

Difference is not significant.

Alternative hypothesis (H_1) : $\mu \neq 16$

calculated value of t :

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n-1}} = \frac{15.8 - 16}{0.5/\sqrt{9}} = -1.2$$

$$\therefore |t| = 1.2$$

$$\text{degrees of freedom } \gamma = n - 1 \\ = 10 - 1 = 9$$

$$\alpha = 5\% = 0.05$$

$$\text{Table value} = 2.262$$

The calculated value $<$ table value

$\therefore$ Accept H_0 .

Thus the difference is not significant.

2. A machinist is expected to make engine parts with an diameter of 1.75 cm. A random sample of 10 parts show a mean diameter of 1.85 cm with standard deviation of 0.1 cm. On the basis of this sample, would you say that the work of the machinist is inferior?

Soln:

Given $n = 10 (< 30)$, $\bar{x} = 1.85$

$\mu = 1.75$ sample S.D. $s = 0.1$

Null hypothesis (H_0): The product is conforming to specification.

ie. $\mu = 1.75$

Alternative hypothesis: (H_1): $\mu \neq 1.75$

$$\text{Test statistic } t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{1.85 - 1.75}{0.1/\sqrt{9}}$$

$$\boxed{\therefore t = 3}$$

$$\text{degrees of freedom } \gamma = n - 1 = 10 - 1 = 9$$

$$\alpha = 5\% = 0.05$$

$$\text{Table value} = 2.26$$

Since, calculated value $>$ table value

$\therefore$ Reject H_0 .

Thus, the work of the machinist is inferior.

3. Sandal powder is packed into packets by a machine. A random sample of 12 packets is drawn and their weights are found to be (in kg) 0.49, 0.48, 0.47, 0.48, 0.49, 0.50, 0.51, 0.49, 0.48, 0.50, 0.51, 0.48. Test if the average packing can be taken 0.5 kg.

Soln:

Given $n = 12$, $\mu = 0.5 \text{ kg}$

$$\bar{x} = \frac{\sum x}{n} = \frac{1}{12} [0.49 + 0.48 + 0.47 + 0.48 + 0.49 + 0.50 + 0.51 + 0.49 + 0.48 + 0.50 + 0.51 + 0.48]$$

$$= \frac{5.88}{12} = 0.49$$

to find Sample S.D (s)

$$s^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2$$

$$= \frac{\sum x^2}{n} - (\bar{x})^2$$

$$= \frac{1}{12} \left[0.49^2 + 0.48^2 + 0.47^2 + 0.48^2 + 0.49^2 + 0.50^2 + 0.51^2 + 0.49^2 + 0.48^2 + 0.50^2 + 0.51^2 + 0.48^2 \right] - (0.49)^2$$

$$= \frac{2.883}{12} - (0.49)^2$$

$$= 0.24025 - 0.2401$$

$$s^2 = 0.00015 \Rightarrow s = 0.01224745$$

Null hypothesis (H_0) : There is no significance difference between the population mean and sample mean. $\mu = 0.5$

Alternative hypothesis (H_1) : $\mu \neq 0.5$

Calculated value of t:

$$\therefore t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{0.49 - 0.5}{0.01224745/\sqrt{11}}$$

$$= -2.708013$$

$$|t| = 2.70813 //$$

degrees of freedom $\gamma^2 = n-1$
 $= 12-1 = 11$

$\alpha = 5\%$

table value $= 2.20$

$\therefore$ calculated value $>$ table value

$\therefore$ Reject H_0

$\therefore$ Average parking cannot be taken to be 0.5 kg.

Type-II

t' test of difference of Means:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \text{ where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

The degrees of freedom $\gamma^2 = n_1 + n_2 - 2$.

Note:

Sample Size n_1 & n_2 ,

means $\bar{x}_1$ & $\bar{x}_2$

S.D $\Rightarrow s_1$ and s_2

①. Two independent samples of size 15 are taken from normal populations. The sample means are 26 and 10 and variances are 7 and 4 respectively. Is the difference between the means significant?

Soln:

Given $n_1 = n_2 = 15$, $\bar{x}_1 = 26$, $\bar{x}_2 = 10$

$$s_1^2 = 7 \text{ and } s_2^2 = 4.$$

Null hypothesis (H_0): There is no significant difference between the two sample means.

$$\mu_1 = \mu_2$$

Alternative hypothesis (H_1): $\mu_1 \neq \mu_2$

Test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{15 \cdot 7 + 15 \cdot 4}{15 + 15 - 2} = 5.89286$$

$$\therefore S = \sqrt{5.89286} = 2.4275$$

$$\therefore t = \frac{26 - 10}{2.4275 \sqrt{\frac{1}{15} + \frac{1}{15}}} = 18.0506 //$$

degrees of freedom $\gamma = n_1 + n_2 - 2$

$$= 15 + 15 - 2 = 28$$

$$\alpha = 5\%$$

$$\text{table value} = 2.05$$

since, the calculated value $>$ table value

$\therefore$ Reject H_0 .

The difference of means is significant.

2. The following table gives the values of protein from Kargayam cow's milk and buffalo's milk. By using t test examine if these differences are significant.

Cow's milk : 1.90 1.95 2 2.02 1.85 1.85

Buffalo's milk : 2.12 2.00 2.20 2.45 2.20 2.10

Soln:

Let x_1 & x_2 be the values of protein from Kargayam cow's milk and buffalo's milk.

Given $n_1 = 6$, $n_2 = 6$

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{11.52}{6} = 1.92$$

$$\bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{13.07}{6} = 2.178$$

x_1	x_2	$(x_1 - \bar{x}_1)$	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)$	$(x_2 - \bar{x}_2)^2$
1.9	2.12	-0.02	0.0004	-0.058	0.003364
1.95	2	0.03	0.0009	-0.178	0.031684
2	2.2	0.08	0.0064	0.022	0.000484
2.02	2.45	0.1	0.01	0.272	0.073984
1.85	2.2	-0.07	0.0049	0.022	0.000484
1.85	2.1	-0.07	0.0049	-0.078	0.006084
			<u>0.0275</u>		
				<u>0.116084</u>	

$$S^2 = \frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$$

$$= \frac{0.0275 + 0.116084}{6 + 6 - 2} = 0.014358$$

$$\therefore S = 0.119825$$

Null hypothesis: (H_0): There is no significant difference between the two sample means.

$$\text{i.e. } \mu_1 = \mu_2$$

Alternative hypothesis: (H_1): $\mu_1 \neq \mu_2$

$$\therefore t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= 1.92 - 2.178$$

$$0.119825 \times \sqrt{\frac{1}{6} + \frac{1}{6}}$$

$$= -3.7293$$

$$\therefore |t| = 3.7293$$

degrees of freedom $\nu = n_1 + n_2 - 2$

$$= 6 + 6 - 2 = 10$$

$$\alpha = 5\% = 0.05$$

$$\text{table value} = 2.33$$

Since, the calculated value $>$ table value

$\therefore$ Reject H_0 .

The difference of means is significant.

3. Sample of two types of electric bulbs are tested for length of life and the following

Type	Sample size	sample mean	S.D
I	8	1234 hrs	36 hrs
II	7	1036 hrs	40 hrs

If the difference in the means sufficient to warrant the type 1 is superior to type 2 regarding the length of life.

Soln:

$n_1 = 8$, $n_2 = 7 (\leq 30)$ small samples

$$\bar{x}_1 = 1234 \quad \bar{x}_2 = 1036 \quad , \quad s_1 = 36, s_2 = 40$$

H₀: The difference in the mean is sufficient to warrant the type 1 is superior to type 2. $[\mu_1 = \mu_2]$

H₁: The difference in the mean is not sufficient to warrant the type 1 is superior to type 2. $[\mu_1 \neq \mu_2]$

calculate
$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where
$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(8)(36)^2 + 7(40)^2}{8+7-2}$$

$$S = \sqrt{\frac{10368 + 11200}{13}}$$

$$S = 40.731.$$

$$\therefore \text{calculated value } t = \frac{1234 - 1036}{40.731 \sqrt{\frac{1}{8} + \frac{1}{7}}}$$

$$= 9.39$$

$$\text{degrees of freedom } \gamma = n_1 + n_2 - 2$$

$$= 8 + 7 - 2 = 13$$

$$\alpha = 5\% = 0.05$$

$$\text{Table value} = 2.16$$

Since, calculated value > table value

$\therefore$ Reject H_0 .

The difference in the mean is not sufficient to warrant the type 1 is superior to type 2.

4. Intelligence test of two groups of Boys and girls gave the following results

group of 12 girls	mean	S.D
	84.11	10

group of 8 boys	Mean	S.D
	81	12

Examine the difference of the means are Significant.

Soln: $n_1 = 12$, $n_2 = 8 \leq (30)$ Small sample

$$\bar{x}_1 = 84, \bar{x}_2 = 81$$

$$s_1 = 10, s_2 = 12$$

H_0 : The difference of the means are Significant

H_1 : The difference of the means are not

Significant.

$$\text{calculated value } \Rightarrow t = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

$$= \sqrt{\frac{12(10)^2 + 8(12)^2}{12 + 8 - 2}} = \sqrt{\frac{1200 + 1152}{18}}$$

$$s = 11.430$$

$$t = \frac{84 - 81}{11.43 \sqrt{\frac{1}{12} + \frac{1}{8}}}$$

$$= \frac{3}{11.43 \sqrt{0.208}} = \frac{3}{5.217}$$

$$t = 0.575$$

degrees of freedom $\gamma = n_1 + n_2 - 2 = 12 + 8 - 2 = 18$

$$\alpha = 5\% = 0.05$$

Table value = 2.101

Since, calculated value $<$ table value

accept H_0 .

The difference of the means are significant.

5. The mean of two samples of signs 9 and 7 are 196.42 and 198.82 respectively. The sum of squares of the deviation from the mean are 26.94 and 18.73 respectively. Can the samples be considered to have come from a normal population of equal means.

Soln:

$$n_1 = 9, n_2 = 7$$

$$\sum (x_1 - \bar{x}_1)^2 = 26.94$$

$$\sum (x_2 - \bar{x}_2)^2 = 18.73$$

$$\bar{x}_1 = 196.42 \quad \bar{x}_2 = 198.82$$

H_0 : The samples he considered to have come from a normal population of equal means

H_1 : The samples not be considered to have come from a normal population of equal means.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$S = \sqrt{\frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}}$$

$$= \sqrt{\frac{26.94 + 18.73}{9 + 7 - 2}}$$

$$= \sqrt{\frac{45.67}{14}} = \sqrt{3.262}$$

$$S = 1.806$$

$$\therefore t = \frac{196.42 - 198.82}{1.806 \sqrt{\frac{1}{9} + \frac{1}{7}}}$$

$$= \frac{-2.4}{1.805 \times 0.503} = -2.643$$

$$= 2.643$$

$$\text{degrees of freedom} = n_1 + n_2 - 2$$

$$= 9 + 7 - 2$$

$$= 14$$

$$\alpha = 5\% = 0.05$$

$$\text{table value} = 2.145$$

Since, Calculated value > Table value

Reject H_0

Thus the samples not be considered to have come from a normal population of equal means.

Cal t - Tab t

0.87 < 2.12

We accept H_0 .

This shows the superiority of diet A over diet B.

F-test.

2 Marks.

1) What are the uses of F-test?

It is used to test whether two independent samples have been drawn from the ^{normal} population with same variances σ^2 .

2) * It is used to test whether the independent estimates of the population variance are homogeneous or not.

Working rule:-

* Write the sample size n_1 and n_2 and variances s_1^2 and s_2^2

* State H_0 and H_1

* Step 3:- Case (1) Cal $F = \frac{s_1^2}{s_2^2}$ $s_1^2 > s_2^2$

Case (2) Cal $F = \frac{s_2^2}{s_1^2}$ $s_2^2 > s_1^2$

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1}$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1}$$

} if given directly

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}$$

$$S_2^2 = \frac{\sum (x - \bar{x})^2}{n_2 - 1}$$

} sum of squares of deviation
if set of values are given

Step 4.

Tabulated Value of F

Case i) If $S_1^2 > S_2^2$

Tab F = $(n_1 - 1, n_2 - 1)$ at 5% LOS

Case ii) If $S_2^2 > S_1^2$

Tab F = $(n_2 - 1, n_1 - 1)$ at 5% LOS

Step 5:-

If Tab F > Cal F

We accept H_0

If Cal F > Tab F

We reject H_0

Q.1

1) Two random samples gave the following results

Sample	Size	Mean	Sum of the squares of deviation from mean
--------	------	------	---

I	10	15	90
II	12	14	108

Test whether the difference between the variances is significant.

F-test

Soln $n_1 = 10$ $n_2 = 12 \leq 30$ (small sample)

$$S_1^2 = \frac{\sum (x_1 - \bar{x})^2}{n_1 - 1} = \frac{90}{10 - 1} = \frac{90}{9} = 10.$$

$$S_2^2 = \frac{\sum (x_2 - \bar{x})^2}{n_2 - 1} = \frac{108}{12 - 1} = \frac{108}{11} = 9.818$$

$H_0 \rightarrow$ The difference between the variances are significant.

$$\text{Cal } F = \frac{S_1^2}{S_2^2} \quad (S_1^2 > S_2^2)$$

$$= \frac{90}{9.818}$$

$$\text{Cal } F = 1.01$$

$$\text{Tab } F = (n_1 - 1, n_2 - 1) \text{ at } 5\% \text{ LOS}$$

$$= (10 - 1, 12 - 1) \text{ at } 5\% \text{ LOS}$$

$$= (9, 11) \text{ at } 5\% \text{ LOS}$$

$$\text{Tab } F = 2.90$$

$$\text{Cal } F \quad \text{Tab } F$$

$$1.01 < 2.90$$

We accept H_0

The difference between the variances are ~~different~~ significant.

- 2) In a sample of 8 observations, the sum of squares of deviation of the sample values from the sample mean was 94.5 and in another sample of 11 observation, it was found to be 101.7

Test whether the difference of variation is significant.

Sln

$$n_1 = 8 \quad n_2 = 11 \leq 30 \text{ (small sample)}$$

$$S_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{94.5}{7} = 13.5$$

$$S_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{101.7}{10} = 10.17$$

H_0 = The difference of variation is significant

H_1 = The difference of variation is insignificant

$$\text{Cal } F = \frac{13.5}{10.17} \quad (S_1^2 > S_2^2)$$

$$\text{Cal } F = 1.261$$

$$\text{Tab } F = (n_1 - 1, n_2 - 1) \text{ at } 5\% \text{ LOS}$$

$$= (7, 10) \text{ at } 5\% \text{ LOS}$$

$$= 3.14$$

Tab F Cal F
3.14

Cal F Tab F
 $1.261 < 3.14$

We accept H_0

The difference of variation is significant.

Note Standard deviation = $\sqrt{\text{Variance}}$

Variance = (Standard deviation)²

times
Pumpkins were grown under two random
Sample of 11 and 9 items shows their
the Sample S.D of the weights
are 0.8 and 0.5 respectively. Assuming
that the weights distributions are
normal. Test the hypothesis that
the true variances are equal.

soln.

$$n_1 = 11 \quad n_2 = 9$$

$$S_1 = S.D = 0.8 \quad S_2 = S.D = 0.5$$

$$S_1^2 = (0.8)^2 \quad S_2^2 = (0.5)^2$$

$$S_1^2 = 0.64 \quad S_2^2 = 0.25$$

H_0 = The ~~true~~ variances are equal

H_1 = The true variances are not equal.

$$S_1^2 = \frac{n_1 S_1^2}{n_1 - 1} = \frac{11 \times (0.64)}{11 - 1} = 0.704$$

$$S_2^2 = \frac{n_2 S_2^2}{n_2 - 1} = \frac{9 \times 0.25}{9 - 1} = 0.281$$

$$Cal F = \frac{S_1^2}{S_2^2} = \frac{0.704}{0.281} = 2.505$$

$$Tab F = (n_1 - 1, n_2 - 1) \text{ at } 5\% LOS$$

$$= (10, 8) \text{ at } 5\% LOS$$

$$Tab F = 3.35$$

Cal F Tab F
2.505 < 3.35

We accept H_0 .

The true variances are equal.

Two independent samples of sizes 6 and 5 from a normal population had the following values of a variable.

Sample 1	18	13	12	15	12	14
Sample 2	16	19	13	16	18	-

No. They estimate that of population variant differs significantly at 5% LOS.

$n_1 = 6$ $n_2 = 5 < 30$ (small sample)

Sln Let Sample 1 = x_1

Sample 2 = x_2

x_1	x_2	$x_1 - \bar{x}_1$ ($x_1 - 14$)	$x_2 - \bar{x}_2$ $x_2 - 16.4$	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)^2$
18	16	4	-0.4	16	0.16
13	19	-1	2.6	1	6.76
12	13	-2	-3.4	4	11.56
15	16	1	-0.4	1	0.16
12	18	-2	1.6	4	2.56
14		0		0	

$\sum x_1 = 84$ $\sum x_2 = 82$
 $n_1 = 6$ $n_2 = 5$
 $\bar{x}_1 = 14$ $\bar{x}_2 = 16.4$

$= 26$ $= 26.2$

Step 2

H_0 = They estimate of population variant differs significantly at 5% LOS.

H_1 = Population variant does not differ significantly at 5% LOS.

Step 3:-

$$S_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{26}{5} = 5.2$$

$$S_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{21.2}{4} = 5.3$$

$$\text{Cal F} = \frac{S_2^2}{S_1^2} = \frac{5.3}{5.2} = 1.01.$$

Step 4:-

$$\text{Tab F} = (n_2 - 1, n_1 - 1)$$

$$= (4, 5) \text{ at } 5\% \text{ LOS}$$

$$\text{Tab F} = 5.19.$$

$$\text{Cal F} \quad \text{Tab F}$$

$$1.01 < 5.19$$

We accept H_0 .

They estimate of population variant differs significantly at 5% LOS.

Two random sample drawn from two normal population have the values as below.

Sample 1: 19 17 16 28 12 23 19 24 26

Sample 2: 28 32 40 37 30 35 40 28 41 45 30 36

Obtain the estimate of variance of the population and test whether the two populations have the same variance.

Soln $n_1 = 9$ $n_2 = 12 \leq 30$ (Small Sample)

x_1	x_2	$x_1 - \bar{x}_1$	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)$	$(x_2 - \bar{x}_2)^2$
19	28	-2.55	6.5	-7.16	51.26
17	32	-4.55	20.7	-3.16	9.98
16	40	-5.55	30.80	4.84	23.42
28	37	6.45	41.6	1.84	3.3856
12	30	-9.55	91.2	-5.16	26.625
23	35	1.45	2.1025	-0.16	0.0256
19	40	-2.55	6.5	4.84	23.425
24	28	2.45	6	-7.16	51.2656
26	41	4.45	19.8	5.84	34.1056
-	45			9.84	96.825
-	30			-5.16	26.625
-	36			0.84	0.7056

$$\sum x_1 = 184 \quad \sum x_2 = 422$$

$$\bar{x}_1 = \frac{184}{9} \quad \bar{x}_2 = \frac{422}{12}$$

$$\bar{x}_1 = 20.44 \quad \bar{x}_2 = 35.16$$

$$\sum (x_1 - \bar{x}_1)^2 = 134.22$$

$$\sum (x_2 - \bar{x}_2)^2 = 347.66$$

$$\begin{aligned}
 S_1 &= \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} \\
 &= \frac{134.22}{9 - 1} \\
 &= \frac{134.22}{8}
 \end{aligned}$$

$$S_1 = 16.77$$

$$\begin{aligned}
 S_2 &= \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} \\
 &= \frac{347.66}{12 - 1} \\
 &= \frac{347.66}{11}
 \end{aligned}$$

$$S_2 = 31.605$$

$H_0 =$ The two populations have the same variant

$H_1 =$ The two populations do not have same variant.

$$\text{Cal } F = \frac{S_2^2}{S_1^2} \quad (S_2^2 > S_1^2)$$

$$\text{Cal } F = \frac{31.605}{16.77} = 1.8846$$

$$\text{Tab } F = (n_2 - 1, n_1 - 1) \text{ at } 5\% \text{ LOS}$$

$$= (11, 8) \text{ at } 5\% \text{ LOS}$$

$$\text{Tab } F = 2.95$$

$$\text{Cal } F \quad \text{Tab } F$$

$$1.8846 < 2.95$$

We accept H_0

The two populations have the same variant.

Type-2:-
Two random samples gave the following data

Sample No	Size	Mean	Variance
1	8	9.6	1.2
2	11	16.5	2.5

Can we conclude that two samples can be drawn from the same normal population

Ans Note:- The normal population have two parameters

i) Mean

ii) Variant

i) Mean:-

i) By Using t-Test (difference of means)

$$t_{\text{alt}} = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$S = \sqrt{\frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}}$$

$$S = \sqrt{\frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}}$$

ii) By Using F-Test

$$F = \frac{S_1^2}{S_2^2}$$

$$S_1^2 = \frac{n_1 S_1^2}{n_1 - 1} \quad S_2^2 = \frac{n_2 S_2^2}{n_2 - 1}$$

$$S_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} \quad S_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1}$$

H_0 = Two samples can be drawn from the same normal population

H_1 = Two samples cannot be drawn from the same normal population

Sln F-test

$$n_1 = 8 \quad n_2 = 11 \leq (\text{small sample})$$

$$s_1^2 = 1.2$$

$$s_2^2 = 2.5$$

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{8 \times 1.2}{8 - 1} = 1.371$$

$$S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{11 \times 2.5}{10} = 2.75$$

$$\text{Cal } F = \frac{S_2^2}{S_1^2} = \frac{2.75}{1.371} = 2.005$$

$$\begin{aligned} \text{Tab } F &= (n_2 - 1, n_1 - 1) \text{ at } 5\% \text{ LOS} \\ &= (10, 7) \text{ at } 5\% \text{ LOS} \end{aligned}$$

$$\text{Tab } F = 3.64$$

Step 1

$$\text{Cal } F \quad \text{Tab } F$$

$$2.005 < 3.64$$

We accept H_0 .

Two samples can be drawn from the same normal population

t-test

$$n_1 = 8$$

$$n_2 = 11$$

$$\bar{x}_1 = 9.6$$

$$\bar{x}_2 = 16.5$$

Variance $s_1^2 = 1.2$

$$s_2^2 = 2.5$$

$$s = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

$$s = \frac{(8 \times 1.2) + (11 \times 2.5)}{8 + 11 - 2}$$

$$s = 1.477$$

$$\text{Calc} = \frac{\bar{x}_1 - \bar{x}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{9.6 - 16.5}{1.477 \sqrt{\frac{1}{8} + \frac{1}{11}}}$$

$$\text{Calc} = |-10.05|$$

$$= 10.05$$

$$\text{Tab } t = (n_1 + n_2 - 2) \text{ at } 5\% \text{ LOS}$$

$$= (8 + 11 - 2)$$

$$\text{Tab } t = 17 \text{ at } 5\% \text{ LOS.}$$

$$F_{\text{tab}} = 2.11$$

$$F_{\text{cal}} > F_{\text{tab}}$$

$$10.05 > 2.11$$

We reject H_0

We accept H_1

Two samples cannot be drawn from the same normal population

Q.10 The nicotine content in two samples of Tobacco are given below

Sample 1:-	21	24	25	26	27	-
Sample 2:-	22	27	28	30	31	36

Can you say that two samples came from the same normal population?

Soln $n_1 = 5$ $n_2 = 6 \leq 30$ (small sample)

F-test

$$S_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1}$$

$$S_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1}$$

x_1	x_2	$x_1 - \bar{x}_1$	$x_2 - \bar{x}_2$	$(x_1 - \bar{x}_1)^2$	$(x_2 - \bar{x}_2)^2$
21	22	-3.6	-7	12.96	49
24	27	-0.6	-2	0.36	4
25	28	0.4	-1	0.16	1
26	30	1.4	1	1.96	1
27	31	2.4	2	5.76	4
	36		7		49

$$\sum x_1 = 123 \quad \sum x_2 = 174$$

$$\bar{x}_1 = \frac{123}{5} \quad \bar{x}_2 = \frac{174}{6}$$

$$\bar{x}_1 = 24.6 \quad \bar{x}_2 = 29$$

$$\sum (x_1 - \bar{x}_1)^2 = 21.2 \quad \sum (x_2 - \bar{x}_2)^2 = 108$$

$$S_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{21.2}{5 - 1} = \frac{21.2}{4} = 5.3$$

$$S_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{108}{6 - 1} = \frac{108}{5} = 21.6$$

H_0 = The two samples came from the same normal population

H_1 = The two samples not came from the same normal population

$$\text{Cal } F = \frac{s_2^2}{s_1^2} = \frac{21.6}{5.3} = 4.075$$

$$\text{Cal } F = 4.075$$

$$\text{Tab } F = (n_2 - 1, n_1 - 1) \text{ at } 5\% \text{ LOS.}$$

$$\text{Tab } F = (5, 4) \text{ at } 5\% \text{ LOS}$$

$$\text{Tab } F = 5.19 \text{ at } 5\% \text{ LOS}$$

$$\text{Cal } F \quad \text{Tab } F$$

$$4.075 < 5.19$$

We accept H_0 .

The two samples came from the same normal population.

T-test $n_1 = 5$ $n_2 = 6 \leq 30$ (Small Sample).

$$\bar{x}_1 = 21.6 \quad \bar{x}_2 = 29$$

$$S = \sqrt{\frac{\sum(x_1 - \bar{x}_1)^2 + \sum(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}}$$

$$= \sqrt{\frac{21.2 + 108}{5 + 6 - 2}}$$

$$S = \sqrt{14.355}$$

$$S = 3.788$$

$$\text{cal } t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{21.6 - 29}{3.788 \sqrt{\frac{1}{5} + \frac{1}{6}}}$$

$$= \frac{21.6 - 29}{3.788 \times 0.605}$$

$$\text{Cal } t = -1.9199$$

$$\text{Cal } t = 1.919$$

$$\text{Tab } t = (n_1 + n_2 - 2) \text{ at } 5\% \text{ LOS}$$

$$= (5 + 6 - 2) \text{ at } 5\% \text{ LOS}$$

$$\text{Tab } t = 9 \text{ at } 5\% \text{ LOS}$$

$$\text{Tab } t = 2.262$$

$$\text{Tab } t \quad \text{Cal } t$$

$$2.262 > 1.919$$

We accept H_0

The two sample came from the same normal population.

$$887.8 = 2$$

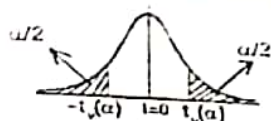
t-test

STATISTICAL TABLES

TABLE II

Critical values of t-Distribution

$$P(t > t_{\alpha}) = \alpha$$



Degrees of freedom (v)	Level of significance (α)				
	0.20	0.10	0.05	0.02	0.01
1	1.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.47	2.015	2.571	3.365	4.022
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
Inf.	1.282	1.645	1.960	2.326	2.576

Note: The above table gives values under both the tails as shown in the diagram. If only one tailed test is used, the significance level given above in the table would be divided by 2, i.e. the second column would give values corresponding to 0.05 instead of 0.10 and the fourth column would give values corresponding to 0.01 instead of 0.02.

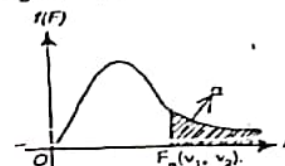
C. Kiran Kumari

STATISTICAL TABLES

TABLE III

Critical Values of F-Distribution
(Right tail areas)

5% and 1% level of significance



The following table provides the values of $F_{\alpha}(v_1, v_2)$. The values of $F_{0.05}(v_1, v_2)$ are given in light face type, while those of $F_{0.01}(v_1, v_2)$ are given in bold face type.

Here, v_1 is the number of degrees of freedom in Nr
 v_2 is the number of degrees of freedom in Dr

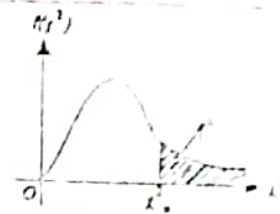
Degrees of Freedom in Dr (v_2)	Degrees of Freedom in Numerator (v_1)											
	1	2	3	4	5	6	7	8	9	10	11	12
1	161 4.052	200 4.999	216 5.403	225 5.625	230 5.764	234 5.859	237 5.928	239 5.981	241 6.022	242 6.055	243 6.082	244 6.106
2	18.51 52.45	19.00 52.00	19.16 51.17	19.25 50.25	19.30 49.30	19.33 48.33	19.36 47.36	19.37 46.37	19.38 45.38	19.39 44.40	19.40 43.41	19.41 42.42
3	10.13 34.12	9.55 30.82	9.28 29.46	9.12 28.71	9.01 28.24	8.94 27.91	8.88 27.67	8.84 27.45	8.81 27.24	8.78 27.03	8.76 26.83	8.74 26.65
4	7.71 21.20	6.94 18.00	6.59 16.69	6.39 15.98	6.26 15.52	6.16 15.21	6.09 14.98	6.04 14.80	6.00 14.66	5.96 14.54	5.93 14.45	5.91 14.37
5	6.61 16.26	5.79 13.27	5.41 12.06	5.19 11.35	5.05 10.97	4.95 10.67	4.88 10.45	4.82 10.25	4.78 10.15	4.74 10.05	4.70 9.96	4.68 9.89
6	5.99 13.74	5.14 10.92	4.76 9.78	4.53 9.15	4.39 8.75	4.28 8.47	4.21 8.26	4.15 8.10	4.10 7.98	4.06 7.88	4.03 7.79	4.00 7.72
7	5.59 12.25	4.74 9.55	4.35 8.45	4.12 7.85	3.97 7.46	3.87 7.15	3.79 6.88	3.73 6.64	3.68 6.47	3.63 6.32	3.60 6.24	3.57 6.17
8	5.32 11.26	4.46 8.85	4.07 7.59	3.84 7.01	3.69 6.63	3.58 6.37	3.50 6.15	3.44 5.93	3.39 5.71	3.34 5.52	3.31 5.44	3.28 5.37
9	5.12 10.55	4.26 8.02	3.86 6.99	3.63 6.42	3.48 6.06	3.37 5.80	3.29 5.52	3.23 5.47	3.18 5.35	3.13 5.26	3.10 5.18	3.07 5.11
10	4.96 10.04	4.10 7.55	3.71 6.55	3.48 5.99	3.33 5.64	3.22 5.39	3.14 5.21	3.07 5.06	3.02 4.95	2.97 4.85	2.94 4.78	2.91 4.71

STATISTICAL TABLES

TABLE : III F DISTRIBUTION (Continued)

Degrees of Freedom in Denominator (v_2)	Degrees of Freedom in Numerator (v_1)											
	1	2	3	4	5	6	7	8	9	10	11	12
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.86	2.82	2.79
12	4.75	3.88	3.49	3.26	3.11	3.00	2.92	2.85	2.80	2.76	2.72	2.69
13	4.67	3.80	3.41	3.18	3.02	2.92	2.84	2.77	2.72	2.67	2.63	2.60
14	4.60	3.74	3.34	3.11	2.96	2.85	2.77	2.70	2.65	2.60	2.56	2.53
15	4.54	3.68	3.28	3.05	2.90	2.79	2.70	2.64	2.59	2.55	2.51	2.48
16	4.49	3.62	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.45	2.42
17	4.45	3.59	3.20	2.96	2.81	2.70	2.62	2.55	2.50	2.45	2.41	2.38
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.37	2.34
19	4.38	3.52	3.13	2.90	2.74	2.63	2.55	2.48	2.43	2.38	2.34	2.31
20	4.35	3.49	3.10	2.87	2.71	2.60	2.52	2.45	2.40	2.35	2.31	2.28
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.28	2.25
22	4.30	3.44	3.05	2.82	2.66	2.55	2.47	2.40	2.35	2.30	2.26	2.23
23	4.28	3.42	3.03	2.80	2.64	2.53	2.45	2.38	2.32	2.28	2.24	2.21
24	4.26	3.40	3.01	2.78	2.62	2.51	2.43	2.36	2.30	2.26	2.22	2.19
25	4.24	3.38	2.99	2.76	2.60	2.49	2.41	2.34	2.28	2.24	2.20	2.17
26	4.22	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.18	2.15

STATISTICAL TABLES

TABLE : IV.
Critical values of χ^2 - Distribution
(Right tail areas)

Degree of freedom (v)	Probability (Level of significance α)						
	0.99	0.95	0.50	0.10	0.05	0.02	0.01
1	0.000157	0.00393	0.455	2.706	3.841	5.214	6.635
2	0.0201	0.103	1.386	4.605	5.991	7.879	9.210
3	0.115	0.352	2.366	6.251	7.815	9.837	11.341
4	0.297	0.711	3.357	7.779	9.488	11.978	13.277
5	0.554	1.145	4.351	9.236	11.070	13.388	15.086
6	0.872	1.635	5.348	10.645	12.592	15.033	16.812
7	1.239	2.167	6.346	12.017	14.067	16.622	18.475
8	1.646	2.733	7.344	13.362	15.507	18.168	20.090
9	2.088	3.325	8.343	14.684	16.919	19.679	21.666
10	2.558	3.940	9.340	15.987	18.307	21.161	23.209
11	3.053	4.575	10.341	17.275	19.675	22.618	24.725
12	3.571	5.226	11.340	18.549	21.026	24.054	26.217
13	4.107	5.892	12.340	19.812	22.362	25.472	27.688
14	4.600	6.571	13.339	21.064	23.685	26.873	29.141
15	4.229	7.261	14.339	22.307	24.996	28.259	30.578
16	5.812	7.962	15.338	23.542	26.296	29.633	32.000
17	6.408	8.672	16.338	24.769	27.587	30.995	33.409
18	7.015	9.390	17.338	25.989	28.869	32.346	34.805
19	7.633	10.117	18.338	27.204	30.144	33.687	36.191
20	8.260	10.851	19.337	28.412	31.410	35.020	37.566
21	8.897	11.591	20.337	29.615	32.671	36.343	38.932
22	9.542	12.338	21.337	30.813	33.924	37.659	40.289
23	10.196	13.091	22.337	32.007	35.172	38.968	41.638
24	10.856	13.848	23.337	32.196	36.415	40.270	42.980
25	11.524	14.611	24.337	33.382	37.652	41.566	44.314
26	12.198	15.379	25.336	34.641	38.885	42.796	45.642
27	12.879	16.151	26.336	35.818	40.113	44.065	46.963
28	13.565	16.923	27.336	36.991	41.337	45.319	48.278
29	14.256	17.708	28.336	38.162	42.557	46.583	49.588
30	14.953	18.493	29.336	39.332	43.773	47.962	50.892

Thus, we conclude that the two samples have not been drawn from the same normal population.

Example 5.4.11. Two random samples gave the following results:

Sample	Size	Mean	Sum of squares of deviations from the mean
1	12	14	108
2	10	15	90

Test whether the samples came from the same population.

Solution . A normal population has two parameters mean μ and variance σ^2 . So we apply t – test for equality of means and F – test for equality of variances.

First we apply F –test:

Step 1: Given $n_1 = 12$, $n_2 = 10$, $\bar{x}_1 = 14$, $\bar{x}_2 = 15$, $\Sigma(x_1 - \bar{x}_1)^2 = 108$, $\Sigma(x_2 - \bar{x}_2)^2 = 90$

$$S_1^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{108}{11} = 9.8182$$

$$S_2^2 = \frac{\Sigma(x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{90}{9} = 10$$

Step 2 : Null hypothesis (H_0) : $\sigma_1^2 = \sigma_2^2$

Alternative hypothesis (H_1) : $\sigma_1^2 \neq \sigma_2^2$.

Step 3: Calculated value of F :

Since $S_1^2 > S_2^2$, then the F –test becomes

$$F = \frac{S_2^2}{S_1^2} = \frac{10}{9.8182} = 1.01852$$

Step 4: Table value of F for $(n_2 - 1, n_1 - 1) = (9, 11)$ degrees of freedom at 5% level of significance = 2.90

Step 5: Conclusion :

At 5% level, the calculated value of $F <$ table value of F .

$\therefore$ Accept H_0 . Thus, the samples came from the populations of equal variance.

To apply t -test:

Step 1:

$$S^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2} = \frac{108 + 90}{12 + 10 - 2} = 9.9$$
$$\therefore S = 3.1464$$

Step 2 : Null hypothesis (H_0) : There is no significance difference between the two sample means. i.e., $\mu_1 = \mu_2$

Alternative hypothesis (H_1) : $\mu_1 \neq \mu_2$.

Step 3: Calculated value of t :

$$\therefore t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{14 - 15}{3.1464 \times \sqrt{\frac{1}{12} + \frac{1}{10}}} = -0.7423$$
$$\therefore |t| = 0.7423$$

Step 4: Table value of t for $\nu = n_1 + n_2 - 2 = 20$ degrees of freedom at 5% level of significance = 2.09

Step 5: Conclusion :

At 5% level, the calculated value of $t <$ table value of t .

$\therefore$ Accept H_0 . Thus, the two samples have been drawn from the same population.

$\therefore$ The two samples have been drawn from same normal population.

Example 5.4.12. Two samples of sizes 9 and 8 gave the sums of squares of deviation from their respective means equal to 160 and 91 respectively. Can they be regarded as drawn from the same normal population?

Nov.'10, Nov.'15

Solution . A normal population has two parameters mean μ and variance σ^2 .

First we apply F -test:

Step 1: Given $n_1 = 9$, $n_2 = 8$, $\Sigma(x_1 - \bar{x}_1)^2 = 160$, $\Sigma(x_2 - \bar{x}_2)^2 = 91$

$$S_1^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{160}{8} = 20$$

$$S_2^2 = \frac{\Sigma(x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{91}{8} = 11.75 \quad [3]$$

Step 2 : Null hypothesis (H_0) : $\sigma_1^2 = \sigma_2^2$

Alternative hypothesis (H_1) : $\sigma_1^2 \neq \sigma_2^2$.

Step 3: Calculated value of F :

Since $S_1^2 > S_2^2$, then the F -test becomes

$$F = \frac{S_1^2}{S_2^2} = \frac{20}{11.75} = 1.7021$$

Step 4: Table value of F for $(n_1 - 1, n_2 - 1) = (8, 7)$ degrees of freedom at 5% level of significance = 3.73

Step 5: Conclusion :

At 5% level, the calculated value of $F <$ table value of F .

$\therefore$ Accept H_0 . Thus, the samples came from the populations of equal variance.

Further, we apply t -test for their means. But, in the given problem the means are not given. So we cannot apply t -test for this problem.

5.6 Part – A : Two Marks Questions

Question 122. Define the 't' statistic.

Apr.'11, Apr.'14

Answer . If $x_1, x_2, \dots, x_n$ be the small samples (i.e., $n < 30$) drawn from the normal population with mean μ and variance S^2 then the sample mean $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ and standard deviation $S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ then the statistic is

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad (\text{or}) \quad \frac{\bar{x} - \mu}{s/\sqrt{n-1}}.$$

Question 123. Write down the 95% and 99% of confidence or fiducial limits of μ . (OR) Give the 95% and 99% confidence interval of the population mean in terms of mean and SD of small sample.

Nov.'15

Answer . 1. 95% confidence limits of mean $\mu = \bar{x} \pm t_{0.05} \frac{S}{\sqrt{n}}$.
2. 99% confidence limits of mean $\mu = \bar{x} \pm t_{0.01} \frac{S}{\sqrt{n}}$.

Question 124. State the properties of t-distribution.

Nov.'10, Apr.'13

Answer . 1. The pdf of t distribution is $f(t) = \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$,

where $t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{\bar{x} - \mu}{s/\sqrt{n-2}}$, $\int_{-\infty}^{\infty} f(t) dt = 1$.

2. The values of t varies from $-\infty$ to ∞ .

3. It is symmetrical about $t = 0$ and mean zero.

4. The variance of the t -distribution is greater than 1 but tends to 1 as $n \rightarrow \infty$. As $\nu \rightarrow \infty$ then t -distribution becomes normal.

5. The variance $= \frac{\nu}{\nu-2}$, if $\nu > 2$ and $\mu_2 > 1$ always.

Question 125. State some applications (or) uses of t -distribution.

Apr.'11

Answer . 1. To test the significance of a single mean.

2. To test the significance difference between the two sample means.

3. To test the significance of the correlation coefficient.
4. To test the significance of observed partial and multiple correlation coefficients.

Question 126. State the assumptions of t -distribution. **Apr.'11**

Answer . 1. The parent population from which the sample is drawn is normal.

2. The sample observations are independent, i.e., the sample is random.
3. The population standard deviation is unknown.
4. Sample size $n < 30$.

Question 127. State the properties of χ^2 distribution. **Nov.'14**

Answer . 1. The sum of independent chi-square variables is also chi-square variable.

2. Chi-square distribution approaches to normal distribution when $n \rightarrow \infty$.
3. The Mgf of chi-square distribution is $(1 - 2t)^{-\nu/2}$.
4. If k is the number of linearly independent constraints then $\nu = n - k$.
5. For $p \times q$ contingency table, the degrees of freedom is $(p-1)(q-1)$.

Question 128. State some applications or Uses of χ^2 distribution **Nov.'10, Nov.'12**

Answer . 1. To test the 'goodness of fit'.

2. To test the 'independence of attributes'.
3. To test the homogeneity of independent estimates of the population variance.
4. To test the hypothetical value of the population variance is σ^2 .

Question 129. What the conditions for application of chi-square test. (OR) State the conditions under which chi-square test of goodness fit is valid.

Nov.'11, Apr.'12, Apr.'15, Nov.'15

Answer . This is an appropriate test for large values of n . For the validity of chi-square test of 'goodness of fit' between the theory and experiment, the following conditions must be satisfied.

1. The sample observations should be independent.
2. Constraints on cell frequencies must be linear.
3. N , the total frequency, should be reasonably large say > 50 .
4. No theoretical frequencies is less than 5. If it is 5, then the application of χ^2 test, it is pooled with the succeeding or preceding so that the pooled frequency is more than 5.

Question 130. A random sample of size 16 has 53 as mean. The sum of squares of the deviations taken from mean is 135. Can this sample be regarded as taken from the populations having 56 as mean?

Solution . Step 1: Given $n = 16$, $\bar{x} = 53$, $\mu = 56$ and $\sum(x - \bar{x})^2 = 135$.

$$\therefore s = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}} = \sqrt{\frac{135}{15}} = 3.$$

Step 2 : Null hypothesis $H_0 : \mu = 56$.

Alternative hypothesis $H_1 : \mu \neq 56$.

Step 3: Calculated value of z

$$\text{Test statistic } z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{53 - 56}{3/\sqrt{16}} = 4$$

Step 4: Tabulated value of z at 5% level of significance = 2.58

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z .

$\therefore$ Accept H_0 . There is no significance difference of the population mean and sample mean.

Question 131. Write the uses of F - distribution.

Nov.'13, Nov.'14, Nov.'15

- Answer .**
1. It is used to test whether two independent samples have been drawn from the normal populations with same variance σ^2
 2. It is used to test whether the two independent estimates of the population variance are homogeneous or not.

Question 132. What are the applications of F - test. Apr.'12

- Answer .**
1. It is used to test whether two independent samples have been drawn from the normal populations with same variance σ^2 .
 2. It is used to test whether the two independent estimates of the population variance are homogeneous or not.

Question 133. Define a 'Attributes'.

Nov.'12

Answer . A mathematical process used to analyze the characteristics of a given population of subjects. Attributes have only two possible ratings (negative or positive) expressed as acceptable or unacceptable, desirable or undesirable, good or bad, etc.

Question 134. A random sample of size 16 has 53 as mean. The sum of squares of the deviation from mean is 135. Can this sample be regarded as taken from the population having 56 as mean? Nov.'12

Solution . Given $n = 16$, $\bar{x} = 53$, $\mu = 56$ and $\sum(x - \bar{x})^2 = 135$.

$$\therefore s = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}} = \sqrt{\frac{135}{15}} = 3.$$

Null hypothesis $H_0 : \mu = 56$. Alternative hypothesis $H_1 : \mu \neq 56$.

$$\text{Test statistic } z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{53 - 56}{3/\sqrt{16}}$$

$$\therefore z = 4$$

Tabulated value of z at 5% level of significance = 2.58

At 5% level, the calculated value of $z <$ tabulated value of z .

$\therefore$ Accept H_0 . There is no significance difference of the population mean and sample mean.

Question 135. Write the probability density function of F – test.
Apr.'13

Answer . The probability distribution of F is given by

$$f(F) = k(F)^{\frac{\nu_1}{2}-1} \left[1 + \frac{\nu_1}{\nu_2} \right]^{-(\nu_1+\nu_2)/2}, \quad 0 \leq F < \infty.$$

Question 136. Define Student's t – test for difference of means of two samples.
Nov.'11

Answer . Consider two independent small samples of sizes n_1 and n_2 . Let $\bar{x}_1$ and $\bar{x}_2$ be the sample means and s_1^2 and s_2^2 be the sample variances. To test the difference between the means are significant using test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}. \quad \text{Where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

The degrees of freedom of this statistic is $\nu = n_1 + n_2 - 2$.

Question 137. Define standard error.
Apr.'11, Apr.'15

Answer . The standard deviation of the sampling distribution of a statistic is known as standard error.

Question 138. Define chi-square test for goodness of fit.
Apr.'11, Apr.'12

Answer . If O_i and E_i ($i = 1, 2, \dots, n$) be the set of observed (experimental) and expected (theoretical) frequencies then χ^2 is defined as

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right]$$